

Class - 03

Measure of Spread

Analyzing Data

Measure Of Spread

We studied that Central Tendency (Mean, Median, Mode) tells us the **Centre**.

But **Spread** tells us **how far the data points are scattered around that center**.

Imagine two cricket players scores an average of 50 runs.

Player A scores 50, 50, 50, 50, 50 (always stable).

Player B: scores 0, 20, 40, 80, 100 (sometimes very low, sometimes very high).

Both have the same mean = 50.

But clearly, **Player B is more inconsistent**, that's because his **spread is larger**.

Measure Of Spread

Now let's talk about Range which is a measure of spread

1. Range :-

Range simply tells us the gap between the **Smallest and the Largest value** in our data.

Formula :-

$$\text{Range} = \text{Largest Value} - \text{Smallest Value}$$

Measure Of Spread

Range :-

Let's see an example for Range,

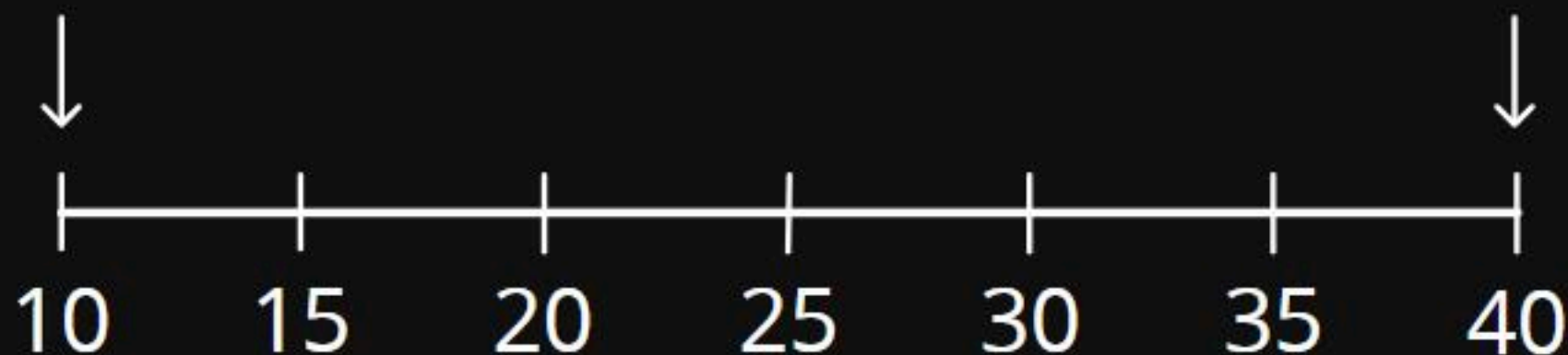
Suppose marks of 5 students and the marks they scored are-

20, 25, 30, 35, 40

- Largest value = 40
- Smallest value = 10
- Range = $40 - 10 = 30$

Smallest value

Largest value



Measure Of Central Tendency

2. Median :-

Median is also used to find the middle point of our data, but here we are just looking for the middle number.

The definition for Median is -

*" The **middle value** when all data is arranged in order. "*

Measure Of Spread

2. INTERQUARTILE RANGE (IQR) :-

After understanding range now we have to focus on **Interquartile Range** for our data.

First, remember

Quartile = Quarter (1/4th)

- The **Interquartile Range (IQR)** tells us how spread out the **middle 50% of the data** is.
- It ignores the lowest values and the highest values and focuses on the central portion of the data.

Measure Of Spread

INTERQUARTILE RANGE (IQR) :-

In IQR, We divide the data into **4 equal parts** and these are called **Quartiles**.

- **Q1** → It is the value at the 25% position (first quartile).
- **Q2** → It is the value at the 50% position (median).
- **Q3** → It is the value at the 75% position (third quartile).
- **Q4** → It is the value at the 100% position (fourth Quartile).

Or in simple words -

- Q1 - 25 percentile.
- Q2 - 50 percentile.
- Q3 - 75 percentile.
- Q4 - 100 percentile.

Measure Of Spread

INTERQUARTILE RANGE (IQR) :-

Now there is a Formula for IQR -

$$\text{IQR} = Q3 - Q1$$

- Here we Subtract Q1 from Q3.

Let's try to understand IQR with an Example

Measure Of Spread

INTERQUARTILE RANGE (IQR) :-

Suppose we have a data -

12,12,13,14,15,17,20,20,28,29,31,35,46,46,48,51,59,60

→ Now we will find the percentile in the data

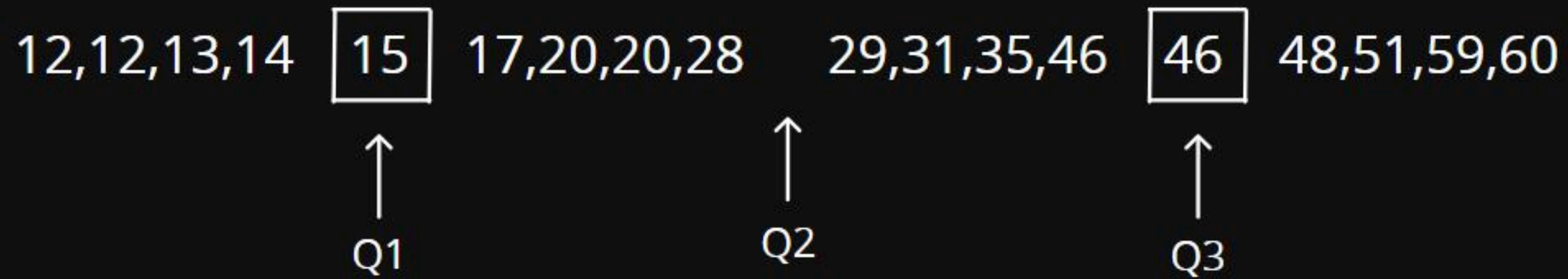
12,12,13,14 **15** 17,20,20,28 29,31,35,46 **46** 48,51,59,60

↑ ↑ ↑

Q1 Q2 Q3

Measure Of Spread

INTERQUARTILE RANGE (IQR) :-



Now we will calculate the IQR,

$$\text{IQR} = Q3 - Q1$$

$$\text{IQR} = 46 - 15 = 21$$

Measure Of Spread

INTERQUARTILE RANGE (IQR) :-

Now the main question comes " Why do we use IQR? "

- IQR tells us how spread out the middle 50% of data is, by subtracting Q1 from Q3.
- IQR **ignores outliers** (extreme values) which can disturb the mean and range.
- IQR gives a **better picture of the central spread** of data.